Unit 5 End-of-Unit Assessment - Form A Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: A parallelogram has base 8 cm, perpendicular height 4 cm, side 6 cm. Find area and identify the unused measure.

Solution / evidence: 32 cm²; slanted side is not the perpendicular height.

Item 2

Prompt: A triangle has area 20 m² and base 10 m. Find its height and check.

Solution / evidence: 4 m; 1/2×10×4=20.

Item 3

Prompt: A trapezoid has bases 4 and 10 cm and height 4 cm. Find area and explain the factor one-half.

Solution / evidence: 28 cm²; two copies make a parallelogram with base 14 and height 4.

Item 4

Prompt: A 12-by-8 cm rectangle has a 3-by-2 cm corner removed. Find remaining area and describe a partition.

Solution / evidence: 90 cm²; subtract 6 from the full area, or use a valid nonoverlapping partition.

Item 5

Prompt: A 4-by-1-by-1/2 cm prism is packed with 1/2 cm edge cubes. Find cube count and volume.

Solution / evidence: 16 cubes, each 1/8 cm³; total 2 cm³.

Item 6

Prompt: Six congruent squares of edge 4 cm are arranged in one straight row. Does this arrangement fold into a cube? Explain why total area alone does not decide validity.

Solution / evidence: No; a row of six squares fails to close without overlap. Validity requires correct face adjacency, not just count; changing square size does not repair the arrangement.

Item 7

Prompt: A prism has 3-4-5 right-triangle ends and length 4 cm. Find surface area, identifying every face.

Solution / evidence: 60 cm²; two triangle ends total 12 and rectangles total 12×4.

Item 8

Prompt: A square pyramid has base edge 8 m and slant height 7 m. Find lateral area for a roof and complete surface area.

Solution / evidence: Lateral 112 m²; complete 176 m²; base excluded for roof covering.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.